

GeoSilo

Compact geometry with explicit precision.

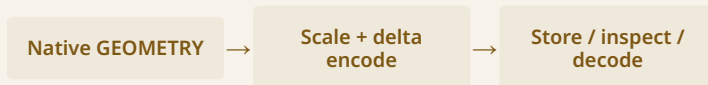


Install & load

```
INSTALL geosilo FROM community;
LOAD geosilo;
```

01 At a glance

Store geometries as scaled integer coordinates with delta encoding. GeoSilo provides a GEOSILO column type, direct measurements and bounds, and explicit encode/decode functions for BLOB pipelines. Choose precision before comparing storage size or query performance.



02 Core SQL surface

```
GEOSILO('EPSG:3857')
```

Typed geometry storage with CRS metadata and automatic coordinate scale.

```
geosilo_encode(geom[, scale])
```

Encode GEOMETRY to BLOB with automatic or explicit integer scale.

```
geosilo_decode(blob)
```

Decode BLOB or GEOSILO to a native GEOMETRY value.

```
geosilo_metadata(blob)
```

Inspect geometry_type, vertex_type, and scale without decoding the geometry.

```
ST_Area / ST_Perimeter
```

Measure GEOSILO polygons directly; results use the coordinate system's units.

```
ST_XMin / ST_XMax
```

Read X bounds; ST_YMin and ST_YMax provide the Y bounds.

```
ST_GeometryType / ST_NPoints
```

Inspect the geometry kind and count of vertices.

03 Keep in mind

Precision follows the scale

Encoding rounds coordinates onto a grid with spacing $1/\text{scale}$. A scale of 100 keeps hundredths of a coordinate unit. Projected-meter defaults use 100; degrees default to 10'. CRS labeling does not transform coordinates.

04 Store a projected polygon

```
CREATE TABLE plots (
  geom GEOSILO('EPSG:3857')
  USING COMPRESSION zstd
);
INSERT INTO plots VALUES (
  'POLYGON((0 0,10 0,10 10,0 10,0 0))'
  ::GEOMETRY
);
```

Inserting GEOMETRY automatically encodes it for the typed column.

05 Inspect and measure directly

```
SELECT ST_Area(geom) AS area,
  ST_Perimeter(geom) AS perimeter,
  geosilo_metadata(geom) AS metadata
FROM plots;
```

Area 100; perimeter 40; integer scale 100. These are planar measurements.

06 Set a scale for a BLOB pipeline

```
WITH encoded AS (
  SELECT geosilo_encode(
    'POINT(1.234 2.345)' ::GEOMETRY, 100
  ) AS blob
)
SELECT geosilo_decode(blob) ::VARCHAR AS wkt
FROM encoded;
```

Returns POINT (1.23 2.35); decoding retains the chosen quantization.

07 In practice

Spatial compatibility

Fixed v1.5 builds merge shared ST_* overloads, so either extension can load first. These examples pass with spatial loaded first on DuckDB 1.5.6. Older GeoSilo d088741 collides with spatial 04270fe; upgrade or use geosilo alone.

[Full docs & examples ↗](#) [Source ↗](#)